Bowen Gu

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RESEARCH INTERESTS

Wireless communications; green IoT; backscatter communications; integrated active-passive communications; mobile edge computing; reconfigurable wireless systems; integrated sensing and communications; AI-enabled and 6G network intelligence.

EDUCATION & APPOINTMENT

2024–present Xinjiang University, School of Computer Science and Technology

2021–2024 Ph.D., Electronic Information Technology, Macau University of Science and Technology

2018–2021 M.Eng., Electronics and Communication Engineering, Chongqing University of Posts and Telecommunications

RESEARCH PROJECTS

2027–2030 PI, NSFC Young Scientists Fund: Reliable Access and Transmission for Backscatter-Assisted Hybrid Active-Passive Communication Systems.

2025–2028 PI, Xinjiang Natural Science Foundation Young Scientists Fund: User Cooperation Mechanism Optimization for Hybrid Active-Passive Communications in Green IoT.

2026–2030 Participant, NSFC Major Program: Information Theory and Controllable Coding of Data Elements.

Participant, Central Government-Guided Local Science and Technology Development Project: Edge Data Governance Innovation Platform Based on Intelligent Signal Detection and Processing.

SELECTED JOURNAL PUBLICATIONS

1. H. Qin, **B. Gu***, D. Li, H. Xie, L. Wang, and X. Tao, “Task Offloading via Hybrid Active-Passive Cooperation: Riding on Active Relaying for Backscatter Communication Enhancement,” *IEEE Internet of Things Journal*, accepted Oct. 2026. DOI: 10.1109/JIOT.2026.3743282.

2. H. Qin, **B. Gu***, et al., “Fairness-Aware Computation Offloading in Wireless-Powered MEC Systems with Cooperative Energy Recycling,” *IEEE Transactions on Mobile Computing*, 2026.

3. G. Li, H. Guo, M. Liu, Z. Wei, **B. Gu**, T. Gu, and D. Ma, “HiRoC: Selective History Routing and Disagreement-Aware Calibration for Multimodal Conversational Emotion Recognition,” *Journal of King Saud University - Computer and Information Sciences*, vol. 38, article 770, 2026.

4. Z. Huang, Y. Xu, **B. Gu**, and H. Zhang, “Resource Management in 6G SAGINs: From Optimization-Based Methods to LLM-Enabled Agentic Decision Making,” *Advanced Information and Communication*, 2026(3):0012.

5. H. Xie, D. Li, **B. Gu**, X. Yu, Y. Xu, and C. Tellambura, “Movable Antenna-Aided Wireless Systems: Concurrent or Cumulative Movement?,” *IEEE Transactions on Communications*, vol. 74, pp. 8829–8842, 2026.

6. Y. He, Z. He, T. Gu, **B. Gu**, Y. Wan, and M. Li, “Multi-Chain of Thought Prompt Learning for Aspect-Based Sentiment Analysis,” *Applied Sciences*, vol. 15, no. 22, article 12225, 2025.

7. **B. Gu**, D. Li, H. Xie, K. Yu, Q. Guan, and Y. Xu, “Exploring Hybrid Active and Passive Multiple Access via Slotted ALOHA-Driven Backscatter Communications,” *IEEE Transactions on Cognitive Communications and Networking*, 2025.

8. **B. Gu**, D. Li, H. Ding, G. Wang, and C. Tellambura, “Breaking the Interference and Fading Gridlock in Backscatter Communications: State-of-the-Art, Design Challenges, and Future Directions,” *IEEE Communications Surveys & Tutorials*, 2025.

9. **B. Gu**, D. Li, Y. Liu, and Y. Xu, “Exploiting Constructive Interference for Backscatter Communication Systems,” *IEEE Transactions on Communications*, 2023.

10. Y. Xu, **B. Gu***, Z. Gao, D. Li, Q. Wu, and C. Yuen, “Applying RIS in Multi-User SWIPT-WPCN Systems: A Robust and Environmentally-Friendly Design,” *IEEE Transactions on Cognitive Communications and Networking*, 2024. ESI Highly Cited Paper.

ACADEMIC SERVICE

2027–2029: Youth Editorial Board Member, *Journal on Communications*.

2026–2028: Youth Editorial Board Member, *China Communications*, *Journal on Internet of Things*, *Journal of Chongqing University of Posts and Telecommunications*, and *Advanced Information and Communication*.

TPC Member: IEEE WCNC 2026; WCSP 2023 and 2024.

Professional memberships: IEEE; IEEE Communications Society; China Institute of Communications; Chinese Institute of Electronics.

Reviewer: IEEE JSAC, TWC, TMC, TCOM, IoTJ, TVT, TCCN, TGCN, WCL, and other SCI journals.

CONFERENCES & ACADEMIC ACTIVITIES

2026.11.06–11.08: Planned participation, 2026 Conference on Frontier Technologies of Information and Communications, Fuzhou.

2026.10.31–11.02: Planned participation, 2026 Internet of Things Technology and Applications Conference, Wuxi.

2026.04.16–04.19: China Electronics Annual Conference, Wuhan.

2025.12.05–12.07: China Communications Society Information Annual Conference, Xi’an.

2025.11.21–11.23: 2025 Conference on Frontier Technologies of Information and Communications, Ningbo.

2025.10.16–10.18: ICCT, Shenyang.

HONORS & AWARDS

2022 Outstanding Master’s Thesis Award of Chongqing; 2021 Outstanding Master’s Thesis Award, CQUPT; 2021–2024 PGS Full Scholarship, Macau University of Science and Technology.